

Ashrith Deshmukh

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EDUCATION

BITS Pilani, KK Birla Goa Campus

B.E. (Hons.) Electrical and Electronics Engineering

05/2024

WORK EXPERIENCE

Data Analyst – Headout

05/2024 – 11/2025 | Bangalore

- Built an **end-to-end competitor intelligence pipeline** using **sentence transformer models** to map competitor catalogs to Headout's; developed a **proxy revenue estimation model** using **review-velocity** as a demand signal, uncovering missing product categories that accounted for ~20% of competitor revenue, directly informing product expansion decisions and serving as a key input for annual and quarterly budget planning.
- Developed a **Retool app** that automated negative review root cause analysis by connecting **Looker views** directly to **Slack**, surfacing complete booking details and investigation context as an instant reply to customer reviews, cutting investigation time by **83% (30 min to 5 min)** for the customer reviews team.
- Built a weekly **Business Intelligence report** using **Looker** tracking key product metrics and growth index scores, enabling managers to identify and take action on underperforming segments directly from the report.
- Owned the end-to-end analytics data stack, building **20+ dbt models** to transform and load data into **BigQuery** and authoring **Looker dashboards** that became the primary reporting layer for marketing growth tracking and day-to-day operational decision making across teams.

Machine Learning Intern – DXC Technology

08/2023 – 12/2023 | Bangalore

- Architected and deployed a **Generative AI health insurance chatbot** from scratch using **Google PaLM** and **LangChain**, enabling users to query complex, jargon-heavy insurance documents in plain English and receive accurate, context-aware responses in real time, eliminating the need for manual policy lookup.
- Implemented a **retrieval-augmented generation (RAG) pipeline** using **FAISS** for **vector similarity search** and **HuggingFace embeddings** for **semantic representation** of insurance documents enabling accurate, document-grounded responses from unstructured PDF data.
- Delivered the **end-to-end solution** within a 5-member **Agile team**, owning the **NLP** and **vector search components** across the full development lifecycle using **PyPDF2**, **LangChain**, and **FAISS**.

Summer Intern – Yashoda Hospitals

06/2022 – 07/2022 | Hyderabad

- Led a team of interns to build an **automated blood report analysis system** using **Tesseract OCR** and **OpenCV** to extract key diagnostic parameters from scanned reports, achieving ~95% extraction accuracy across 100 real hospital reports.
- Designed an **automated abnormality flagging pipeline** that compared extracted blood parameters against **clinical reference ranges**, storing results in a database and surfacing flagged anomalies in a doctor-facing app for faster review.

PROJECTS

Customer Churn Prediction with A/B Test-Driven Feature Validation [GitHub](#)

02/2026

- Built an end-to-end **churn prediction pipeline** using **XGBoost** and **LightGBM** with **Optuna** hyperparameter tuning on the Kaggle Telco dataset, achieving **AUC-ROC of 0.84**, with a **SHAP explainability** layer that surfaces the top 3 churn drivers per customer, directly mirroring how retention teams at product companies consume churn scores in production
- Designed a rigorous **A/B testing framework** from scratch using **scipy.stats** and **statsmodels**, running **two-proportion z-tests** and **Mann-Whitney U tests** with upfront **power analysis** to validate whether engineered features like rolling engagement score and tenure-to-spend ratio produced **statistically significant lift** over baseline
- Deployed the full scoring pipeline as a **FastAPI** and **Docker** REST API backed by a **Streamlit** dashboard that accepts a raw customer CSV and returns churn probabilities, SHAP explanations, and a prioritised risk intervention tier in **under 2 seconds**

LLM-Powered SQL Agent with Self-Healing Query Execution and Observability [GitHub](#)

04/2026

- Built a **natural language to SQL agent** using **LangChain** and **GPT2-medium** over a multi-table e-commerce database, implementing a **self-healing execution loop** that catches SQL runtime errors, passes the trace back to the LLM with a structured correction prompt, and retries autonomously, reducing **query failure rate by 70%** compared to single-pass generation
- Integrated **MLflow** for full **prompt and response observability**, logging token usage, latency, and accuracy scores against a **50-question ground-truth test set**, alongside a **query confidence scoring module** that routes low-confidence outputs to a human review queue before execution, demonstrating **production-grade LLM safety** thinking.
- Containerised the full system with **Docker Compose**, exposing a **Streamlit** chat interface and a live **MLflow tracking dashboard** accessible via a single command, with end-to-end inference benchmarked consistently at **under 3 seconds per query**.

TECHNICAL SKILLS

Python, SQL, TensorFlow, PyTorch, dbt, BigQuery, Looker, Docker, FastAPI, LangChain, Scikit-learn, XGBoost, OpenCV, LoRA Fine-tuning, Generative AI, RAG, MLOps, Git, PySpark, Excel, Sentence Transformers, Agentic AI, Airflow, GCP, Retool, Google PaLM, Scipy, Statsmodels, MLflow, NumPy, Pandas, SHAP, Optuna, LightGBM, Matplotlib

CERTIFICATIONS

Deep Learning Specialization – DeepLearning.AI

12/2023 – 12/2023

Machine Learning Specialization – Coursera

09/2023 – 09/2023